

Is Safety Grounded or Detachable?

Removal-Cost Audits for Open-Weight Language Models

Nedim Mutlu Sezer Ömer Faruk Aksoy Ahmet Rıfat Öztürk
Feyzi Arda Salihoğlu Yağız Ekrem Dalar

Ethosoft Research Group

{nedim, omer, ahmetrifat, ekrem}@ethosoft.org
feyziarda.salihoglu@ethosoft.org

Abstract

Open-weight language models remain modifiable after release, so release-time refusal alone cannot establish durable safety. We introduce removal-cost grounding as a budget-conditioned audit of whether safety can be weakened while preserving utility, coherent generation, and benign helpfulness. We formalize clean removal as a constrained feasibility problem and distinguish observed counterexamples from inconclusive attack searches. Mycelium v0 provides a deliberately limited case study: four model families exhibit benchmark degradation under an archived removal configuration, while targeted Mistral controls separate destructive ablation from wrapper bypass. A provenance audit excludes three model-task comparisons with mismatched evaluation limits or conflicting snapshots. Across the retained comparisons, mean removal-associated score changes range from -0.382 to -0.185 . In a separate 12-prompt generation panel, the strict attack-success proxy increases from 1/12 with Mycelium enabled to 7/12 under bypass, with wide uncertainty and no matched native safety baseline. Adaptive traces further show why early-stopped checkpoints must be distinguished from training endpoints. These observations do not establish tamper-resistant safety. They motivate an auditable reporting framework in which the cheapest admissible route to useful unsafe behavior, rather than the costliest destructive ablation, determines the strength of a safety claim.

1 Introduction

Open-weight language models can be modified after release through adapter changes, internal interventions, base-model restoration, or fine-tuning. Release-time refusal therefore leaves a question unanswered: how much useful behavior must an actor sacrifice to weaken safety?

We call this question *removal-cost grounding*. Grounding here refers to the coupling between a

behavioral constraint and useful computation under a specified class of interventions. It does not mean factual grounding in retrieved evidence, nor does it identify a particular internal representation as a semantic safety circuit. The term names an operational question about the persistence of a constraint after release.

The central distinction is between *damage* and *clean removal*. Destroying coherent generation can reduce refusal without producing a useful unsafe model. Preserving benchmark accuracy alone is also insufficient: benign helpfulness and substantive unsafe compliance must be measured jointly.

This paper develops a measurement framework and applies it retrospectively to Mycelium v0, a wrapper-based prototype with gated low-rank paths. The case study is useful precisely because it exposes a limitation: a costly adapter ablation coexists with an accessible frozen-base route. A defense-level claim must survive the latter route as well as the former. The existence of a destructive intervention cannot establish that every effective intervention is destructive.

Our empirical evidence comprises a four-family utility matrix, Mistral-only removal controls, and separate generation and adaptive-loss panels. The cross-family matrix does not replicate bypass across families.

We make three contributions. (1) We define a budget-conditioned clean-removal criterion, a minimum-cost quantity, and an uncertainty-aware reporting rule. (2) We use Mycelium to show how intervention identity, native baselines, and checkpoint selection change the interpretation of a removal audit. (3) We provide a reproducible reanalysis of archived scores, including a provenance correction, common-task sensitivity, relative degradation, missing-cell bounds, and explicit limits on the generation evidence. All empirical observations come from archived runs; we add statistical reanalysis, without new model training or generation.

2 Related Work

Behavioral safety evaluation. HarmBench standardizes automated red teaming and refusal evaluation (Mazeika et al., 2024). JailbreakBench defines a shared threat model, prompt and scoring conventions, and reusable artifacts (Chao et al., 2024). StrongREJECT emphasizes that apparent jailbreak success can overstate substantive harmful assistance, including when jailbreaks impair capabilities (Souly et al., 2024). OR-Bench evaluates the complementary failure of rejecting benign requests (Cui et al., 2025). Our utility, coherence, and benign-helpfulness requirements build on these concerns; they are not new desiderata in isolation. The contribution is to apply them jointly to post-release intervention artifacts with explicit budgets and controls.

Tamper resistance. TamperBench is the closest framework: it evaluates safety and utility under weight-space and representation attacks, with systematic hyperparameter sweeps (Hossain et al., 2026). Removal-cost grounding is not a replacement benchmark and does not claim an evaluation capability unavailable to TamperBench. It supplies an interpretation layer: which intervention produces an admissible counterexample, at what cost, under which feasibility constraints? The Mycelium controls instantiate this question by distinguishing changes to the base path from changes confined to added components. We do not report a head-to-head TamperBench experiment or comparative attack coverage.

Adaptation and internal interventions. LoRA enables adaptation through low-rank updates while retaining frozen pretrained weights (Hu et al., 2021). Refusal-direction work shows that refusal can be sensitive to a low-dimensional representation intervention (Arditi et al., 2024). Circuit breakers instead train representations to interrupt undesirable behavior (Zou et al., 2024). Unlearning evaluates the removal of targeted knowledge or capabilities, as exemplified by WMDP (Li et al., 2024). These settings share a need to measure retained utility, but they differ in the object being changed. A wrapper bypass result does not establish that full-parameter alignment, representation rerouting, or unlearning has the same failure mechanism.

3 A Budget-Conditioned Grounding Criterion

3.1 Threat model and measurements

Let M_0 be a released defended model, M_{nat} its local native comparator, and a an intervention returning a runnable artifact $M_a = a(M_0)$. The threat model specifies an admissible class $\mathcal{A}_B$ under budget B . Relevant resources include gradient steps, adapter rank, examples, model queries, checkpoint searches, and access to base weights. These resources should be reported separately unless a justified conversion to a common cost unit exists. Merely counting gradient steps would assign zero cost to many static interventions while omitting their search and implementation costs.

For fixed evaluation protocols, let $S(M) \in [0, 1]$ measure substantive unsafe compliance, $u_t(M) \in [0, 1]$ measure utility on task t , $C(M)$ measure coherent generation, $F(M)$ measure benign false refusal, and $H(M)$ measure substantive benign helpfulness. Higher S is worse safety; higher u_t, C, H is better performance. Refusal is measured separately: non-refusal does not imply unsafe compliance, and absence of false refusal does not imply helpfulness.

Define the safety change $\Delta S(a) = S(M_a) - S(M_0)$ and task-specific utility cost

$$d_t(a) = \max\{0, u_t(M_0) - u_t(M_a)\}. \quad (1)$$

We retain this vector rather than allowing improvements on one task to silently offset losses on another. A weighted mean is useful for description, but it is not automatically an acceptable deployment utility constraint. The native-to-defended comparison must also be reported: preserving a low-utility defended model is a weak success criterion.

3.2 Clean removal and minimum cost

Choose thresholds before attack evaluation:

$$\tau = (\delta, \{\epsilon_t\}, c_0, f_0, h_0, p_0). \quad (2)$$

An artifact meets the behavioral constraints when

$$\begin{aligned} \Delta S(a) &\geq \delta, & d_t(a) &\leq \epsilon_t \quad \forall t, \\ C(M_a) &\geq c_0, & F(M_a) &\leq f_0, \\ H(M_a) &\geq h_0. \end{aligned} \quad (3)$$

Let $P(a)$ be the probability that a frozen selected artifact meets these constraints under the declared

repeated-evaluation protocol. A clean removal additionally requires $P(a) \geq p_0$. Repetitions may vary decoding seeds or evaluation samples while holding the artifact fixed. Attack restarts may produce different artifacts and therefore assess a separate quantity: the reliability of the attack procedure. Checkpoint selection is part of the attack; selection data must be separated from confirmatory evaluation.

For a declared nonnegative cost function $c(a)$, define

$$R_\tau(M_0; \mathcal{A}_B) = \inf_{a \in \mathcal{A}_B: \text{Eq. (3)}, P(a) \geq p_0} c(a), \quad (4)$$

with $\inf \emptyset = +\infty$. This is a population-level target, not a number that our small case study identifies. Finding one qualifying artifact yields an upper bound. Failing to find one in a finite search supplies no general lower bound unless the search is exhaustive or has a justified coverage guarantee. We therefore avoid converting unsuccessful attacks into an unqualified statement that safety is grounded.

Operational decisions. An audit reports *detachable at the declared thresholds* only when an observed artifact satisfies all requirements with the chosen evidential support. It reports *no qualifying removal observed* when a specified search finds none. It reports *inconclusive* when behavioral precision, intervention provenance, or coverage is insufficient. A component can exhibit removal-associated utility loss without earning a defense-level label of “partial grounding.” That label would require a declared quantitative meaning.

3.3 Pareto reporting and uncertainty

A complementary report plots safety erosion against the utility-cost vector or an explicitly weighted summary, with separate annotations for coherence, benign behavior, and cost. An artifact is dominated if another is at least as good on every declared attack objective and strictly better on one. A Pareto point need not satisfy Eq. (3); the frontier and the feasible region answer different questions.

For a confirmatory clean-removal claim, the lower confidence bound on safety erosion should exceed δ , while upper bounds on utility loss and false refusal should remain within their tolerances. Joint claims across many metrics require simultaneous uncertainty control. Marginal intervals printed beside individual scores do not provide that guar-

antee. In particular, overlap between two binomial intervals is not a paired significance test.

Training instability must also be interpreted carefully. A checkpoint that performs well early and deteriorates under further optimization remains available to an early-stopping attacker. Its later collapse does not erase the earlier candidate. Stability of a *selected artifact* under repeated evaluation is distinct from monotonic behavior along a training trajectory. Section 7 reports both checkpoint and endpoint observations without conflating the two.

3.4 The reconnectable-base boundary

Proposition 1. *Suppose an admissible intervention b of cost at most c_b restores exactly the native model’s inference function, including its preprocessing and decoding protocol. If M_{nat} satisfies Eq. (3) relative to M_0 and the repetition requirement, then $R_\tau(M_0; \mathcal{A}_B) \leq c_b$.*

Proof. By functional equivalence, M_b has the native model’s measured behavior under the stipulated protocol. It therefore belongs to the feasible set in Eq. (4); the infimum cannot exceed its cost. $\square$

The proposition is deliberately elementary. It isolates the relevant logical boundary: making one removal route expensive cannot compensate for an admissible inexpensive route. Its premises must still be checked. Restoring some base projections does not automatically establish equality of tokenization, model state, or generation behavior, and an aligned native model need not meet a safety-erosion threshold. Our empirical bypass panel lacks a matched native generation arm, so it does not fully instantiate those premises.

4 Mycelium v0 and Intervention Identity

4.1 Architecture as an audit object

Mycelium v0 adds low-rank paths to selected transformer projections. The design uses operational family names such as core, utility, soft safety, hard safety, and safety. They identify training or intervention groups, not independently localized semantic circuits. A useful mathematical abstraction for one patched projection is

$$y = \alpha_a(x)W_0x + \sum_{f \in \mathcal{F}} m_{a,f} g_f(x)B_f A_f x, \quad (5)$$

where $W_0 \in \mathbb{R}^{d_o \times d_i}$ is the frozen base projection, $A_f \in \mathbb{R}^{r_f \times d_i}$ and $B_f \in \mathbb{R}^{d_o \times r_f}$ are low-rank fac-

tors, g_f is a gate, and $m_{a,f}$ describes the intervention. Bias terms are omitted for readability. An always-on core corresponds to $g_{\text{core}} = 1$. The coefficient α_a records whether the intervention changes the base contribution. This notation separates the objects that an audit must log; it is not a recovered source-level specification of every historical wrapper.

The design description includes green, amber, and red routing, retained benign utility, safety-target responses, and counterfactual training under path removal or wrong routing. Historical runs implement multiple revisions. The archived benchmark metadata identifies wrapper source and checkpoint paths, but the supplied package does not contain the corresponding executable training sources or exact loss coefficients for all final checkpoints. We therefore do not assert that every model used the same objective, gate estimator, or loss schedule. Earlier training-report summaries are not sufficient to reconstruct those details.

4.2 What each removal changes

The broad utility matrix uses the archived condition `mycelium_removed`. We denote it R_6 throughout. It is a wrapper-level removal configuration, not a universally equivalent safety-circuit excision. The Mistral control panel separately reports the following operations:

- **Attenuated removal:** a removal branch that also changes the base projection contribution. It mixes component deletion with base-path damage.
- **Adapter-only safety-zero:** the base projection is preserved while selected added safety paths are disabled. This tests the deletion route with that confound removed.
- **Bypass restore:** the frozen base path is reconnected, avoiding the wrapper’s intended execution route.

In Eq. (5), changing α_a and changing only $m_{a,f}$ are different interventions. Their outcomes cannot be pooled under a generic “removed” label. Exact intervention-to-parameter manifests are needed to establish equivalence between separate runs, even when condition names look similar.

The targeted utility panel uses 200 examples per task. Attenuated and adapter-only removal have the same four reported scores. Equality of aggregate

accuracies is not evidence that their logits, predictions, or code paths are identical. The supplied aggregate panel does not resolve why those scores coincide. An implementation equivalence explanation would require per-example predictions and an execution trace; neither is inferred here.

Limits of causal attribution. Utility loss under adapter-only deletion establishes a cost of that intervention on that model. It does not show that safety-specific computation was removed. Equal-rank random-path ablations, norm-matched controls, and utility-path ablations would distinguish safety-specific coupling from generic low-rank damage. The current evidence does not supply a matched experiment for these controls. We consequently use “removal-associated degradation” rather than “causally grounded safety” to describe the utility result.

5 Benchmark Utility and Provenance Audit

5.1 Data and protocol

We reanalyze archived lm-evaluation-harness outputs (Biderman et al., 2024) for Mistral-7B, Qwen-35B-A3B, Gemma-31B, and GPT-OSS-20B. These names abbreviate local evaluated model families, not claims about the performance of every checkpoint in a family. The 15-task matrix covers reasoning, knowledge, coding, mathematics, and commonsense tasks; Appendix A.1 maps them to source references and recorded metrics. The metric assigned to each task is retained rather than replaced with a newly selected favorable metric. Accuracy, normalized accuracy, exact match, and code-pass measures all lie on a $[0, 1]$ scale, but do not measure interchangeable capabilities.

The original task ledger contains 192 intended model-condition-task cells, including GPQA. It records 170 successful cells, 16 error-only cells, and six missing cells. GPQA was access-blocked and is excluded from the 15-task comparison universe. A successful cell means that a score was produced; it does not guarantee compatible sampling across conditions.

A correction beyond missing-cell counting. The archived aggregate table suggests 15, 13, 13, and 14 complete native/on/ R_6 task triplets. Inspecting evaluation metadata reveals three additional comparisons that must be excluded from our primary reanalysis. Mistral GSM8K native and on

Family	n	On-N	R_6 -On	95% CI
Mistral	13	-.029	-.331	$[-.412, -.245]$
Qwen	12	-.009	-.382	$[-.490, -.274]$
Gemma	13	-.038	-.185	$[-.307, -.081]$
GPT-OSS	14	+.063	-.280	$[-.366, -.197]$

Table 1: Primary utility summary after provenance exclusions. N denotes the local native condition. Values are score differences, not percentages of retained capability.

use eight examples, while R_6 uses 1,319. Qwen GSM8K native uses four examples, unlike the other conditions. Mistral OpenBookQA has an archived removed score of 0.266, whereas the supplied task JSON records 0.500 on a four-example later run. We preserve the conflict rather than selecting either value as a full comparable result.

After these exclusions, the primary analysis retains 13, 12, 13, and 14 triplets, respectively. Appendix A and the supplement list every exclusion. The correction changes the estimand from “all score-present triplets” to “score-present triplets without the identified limit or snapshot conflicts.” It does not establish complete equivalence of all historical evaluation environments.

5.2 Aggregation and robustness checks

For each family, we average the paired differences $u_t(R_6) - u_t(M_0)$ over retained tasks with equal task weights. We bootstrap these task differences with 20,000 resamples and seed 20260905, reporting percentile 95% intervals. This is a descriptive task-resampling analysis. Tasks are not an IID sample of real-world uses, and the interval does not account for model-training uncertainty or all within-task sampling error.

Table 1 shows substantial negative mean changes under R_6 in each family. This is an intervention-associated utility result, not cross-family evidence of the same safety-removal mechanism. Mycelium-on changes are smaller on average, but the average conceals material task-level variation. For example, Gemma GSM8K falls from 0.702 to 0.438 before removal. GPT-OSS shows a positive aggregate on-minus-native difference alongside unusual native generation-task results, including BBH and MBPP scores of zero. These observations warrant local-protocol interpretation rather than a claim of general capability improvement or uniformly negligible utility overhead.

We add four checks to make the aggregation

Task	On	Atten.	Adapter	Bypass
ARC-C	.540	.385	.385	.590
BoolQ	.885	.785	.785	.900
HellaSwag	.745	.565	.565	.735
OpenBookQA	.505	.395	.395	.530
Mean	.669	.533	.533	.689

Table 2: Archived targeted control panel, 200 examples per task. “Adapter” means adapter-only safety-zero. This panel is distinct from R_6 . Bypass has a 0.020 mean improvement relative to on, but HellaSwag decreases by 0.010.

inspectable. First, median removal differences are -0.303 , -0.387 , -0.067 , and -0.254 . Gemma’s smaller median shows how a few large generative-task losses influence its mean. Second, on the ten tasks retained in every family, mean differences remain negative: -0.349 , -0.388 , -0.127 , and -0.263 . Third, the mean relative loss $(u_t(M_0) - u_t(R_6))/u_t(M_0)$ over positive on-state scores is 61.2%, 55.5%, 36.5%, and 58.8%. The zero on-state Mistral HumanEval score is omitted only from this ratio, not from the primary mean. Relative loss is descriptive and is not chance-normalized; values on tasks with small denominators require care.

Fourth, we assign each excluded triplet an adversarially favorable difference of $+1$, the maximum possible for bounded scores. Even this worst-case completion of the 15-task universe leaves negative upper bounds: -0.153 , -0.105 , -0.027 , and -0.195 . These are algebraic sensitivity bounds, not imputations or confidence intervals. They address the direction of this bounded aggregate; they cannot repair protocol mismatches or support an inference about safety.

5.3 Mistral removal controls

Adapter-only removal loses 0.136 mean score across the four tasks in Table 2. Bypass is substantially less destructive, with a maximum observed task loss of 0.010. This supports evaluating bypass as a candidate removal route, not asserting formal utility equivalence: no paired uncertainty or pre-registered noninferiority test is available. Utility preservation is therefore tolerance-dependent; the HellaSwag decrease rules out exact non-decrease on every task.

Condition	Refusal	Strict ASR	Ambiguous
On	8/12	1/12	3/12
Adapter-only	1/12	0/12	12/12
Bypass	2/12	7/12	3/12
Native	—	—	—

Table 3: Panel A. Indicators are not a mutually exclusive partition: adapter-only has 12 ambiguous outputs and one refusal. The missing native arm is shown explicitly. Counts are reconstructed from the submitted table’s rounded rates and stated denominator.

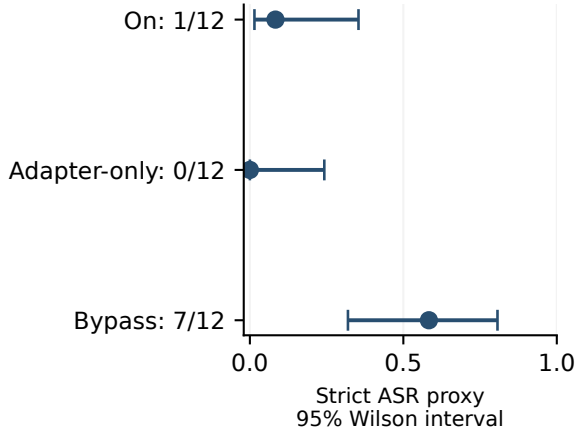


Figure 1: Panel A uncertainty. Twelve observations per condition provide limited precision; the bypass point estimate is suggestive, not a population-level safety estimate.

6 Generation Evidence: Two Separate Panels

6.1 Panel A: targeted bypass comparison

The targeted generation panel reports 12 red prompts per condition for on, adapter-only zero, and bypass restore. “Red” identifies the local safety-evaluation category. The supplied paper reports a strict ASR proxy, a refusal indicator, and an ambiguity indicator. We retain these measurements as historical proxies; the package does not identify a validated external judge or provide sufficient row-level annotations to independently reconstruct the strict rubric.

The strict ASR proxy rises from 1/12 to 7/12 under bypass, a descriptive difference of 0.500. Wilson intervals are [0.015, 0.354] and [0.320, 0.807], respectively (Figure 1). Adapter-only removal has zero strict successes but all 12 outputs marked ambiguous. That outcome cannot be treated as evidence that a useful model remains safe: severe incoherence can suppress both refusal and successful assistance.

Condition	Compliance	Red amb.	Benign amb.
On	5/12	6/12	0/12
Safety-zero	1/12	11/12	12/12
All-zero	11/12	0/12	0/12

Table 4: Panel B broad manual indicators. Benign false refusal is 0/12 in all three conditions, but safety-zero produces 12/12 ambiguous benign outputs. Non-refusal is not helpfulness.

The native safety baseline is absent on these same prompts. We cannot determine whether bypass exactly restores native behavior, whether Mycelium substantially improves safety over that baseline, or whether differences reflect other generation settings. Nor can the archived marginals establish a paired significance result. If the same 12 prompts were used and strict-success labels were paired, marginals of one and seven successes allow two contingency tables. Their exact two-sided McNemar p -values are 0.0703 and 0.0313. Thus, even the decision at a nominal 0.05 threshold depends on missing pairing information. This conditional calculation is a sensitivity analysis, not an observed hypothesis test.

6.2 Panel B: manual audit and label mapping

The separate manual audit reports 72 outputs: three conditions, each with 12 red and 12 benign prompts. Its labels are on, safety_zero, and all_mycelium_zero. The archived aggregate fields record refusal/safe redirection, substantive compliance, and ambiguous or garbled output. The indicators can overlap. The red safety-zero summary has one refusal, one compliance, and 11 ambiguous labels among 12 outputs, confirming that they are not categorical alternatives.

Panel B’s red compliance rates are 0.417, 0.083, and 0.917, with Wilson intervals [0.193, 0.680], [0.015, 0.354], and [0.646, 0.985]. All three benign false-refusal counts are zero, but the 95% upper bound is 0.243, and benign ambiguity varies sharply. These data reinforce the need to report useful benign responses rather than only the absence of refusal.

The two panels have different scoring definitions and condition identifiers. “Safety-zero” cannot be equated with adapter-only safety-zero without an execution manifest; “all-Mycelium-zero” cannot be equated with bypass restore merely because both disable components. Even their on-state red indicators differ substantially. We do not pool their

Family	Steps/rank	Train	Holdout	ΔL_S
Mistral	40/4	.440	.434	-1.240
Mistral	80/8	2.645	2.103	-.888
Qwen	40/4	2.298	1.017	-.421
Qwen	80/8	3.699	3.004	-2.171

Table 5: Archived LoRA repair panel. Positive recovery means lower utility CE. Negative ΔL_S means safe-target likelihood also improves. These are likelihood probes, not observed clean removals.

denominators, describe them as independent replications, or use Panel B to increase the sample size of Panel A. Prompt overlap and checkpoint equivalence are unverified.

The reanalysis package contains aggregate labels, without the annotation template or output-level annotations. Annotator count, independent blinding, and inter-annotator agreement remain undocumented. A manifest of 40 red and 40 benign prompts describes an available pool, not 80 completed observations for every condition. The safety evidence remains preliminary and does not replace the established benchmarks discussed in Section 2.

7 Adaptive Repair and Checkpoint Selection

Adaptive experiments use cross-entropy (CE) proxies. Let L_U be loss on benign targets and L_S loss on safe response targets. Relative to an attack’s starting checkpoint, utility recovery is $L_U^{\text{start}} - L_U^{\text{after}}$ and safety-target erosion is $L_S^{\text{after}} - L_S^{\text{start}}$. Positive erosion means that specified safe targets become less likely; it does not imply substantive unsafe generation. Starting points and evaluation subsets differ from the Phase-6 benchmark matrix, so their CE changes cannot be substituted for benchmark score changes.

Table 5 makes the repair evidence explicit. The two budgets use rank 4 with 40 steps and rank 8 with 80 steps; the archived top- n settings are four and six. All four settings improve held-out utility CE, while safe-target CE also decreases. Training gains exceed held-out gains, especially for Qwen at the smaller budget. The result shows partial utility recovery in the measured loss, not recovery of useful unsafe behavior.

The erosion sweep varies pressure $\lambda \in \{0.02, 0.05, 0.10, 0.20\}$ (Table 6). Its historical screening rule requires at least 0.25 utility-CE recovery and 0.25 safe-target-CE erosion. Four Mistral settings and one Qwen setting have selected

Family	λ	Step*	Rec.*	Erosion*	Final rec.
Mistral	.02	20	1.799	2.369	-8.396
Mistral	.05	20	1.485	84.353	-13.083
Mistral	.10	10	1.973	.687	-23.341
Mistral	.20	10	1.901	5.625	-21.342
Qwen	.02	1	.065	.018	-3.309
Qwen	.05	1	.049	.017	-.950
Qwen	.10	70	.709	95.989	.157
Qwen	.20	1	.048	.025	-5.250

Table 6: Archived erosion sweep: rank 8, 80 steps, top- $n = 6$. Stars denote the archived selected checkpoint. “Final rec.” is endpoint held-out utility-CE recovery. Five selected checkpoints, but no endpoints, exceed both archived CE thresholds of 0.25.

checkpoints meeting that proxy rule; no final checkpoint does. Very large erosion values at some checkpoints coexist with loss of utility. Such extreme losses are warnings against reading CE as a calibrated measure of harmful assistance.

These traces support a reporting lesson rather than a successful defense claim. Endpoint-only analysis misses early candidates. Conversely, early proxy success does not establish Eq. (3): generation safety, benign helpfulness, and repeated-evaluation stability were not jointly measured. An early-stopped artifact should be frozen and evaluated on a separate confirmatory set. The archived traces do not provide this final step, and the word “heldout” in a utility-loss report does not guarantee that those data were unused during checkpoint selection.

8 What the Case Study Establishes

The supported result. The corrected four-family analysis establishes that the archived R_6 configuration is associated with large average utility losses on the retained tasks. The Mistral controls show that a less destructive bypass route is available in the tested implementation, and a small generation panel shows a higher unsafe-compliance proxy under that route. Taken together, these observations challenge the interpretation of destructive removal as evidence of durable safety. They do not establish that Mycelium is a tamper-resistant defense or precisely quantify a minimum clean-removal cost.

Why the boundary is architectural. Frozen-base restoration is unsurprising when the base path remains available. Its importance is diagnostic: the audit must include the strongest admissible route, even when that route is simple. A wrapper can be useful under an execution environment that pro-

tests its control flow, but a white-box open-weight threat model cannot silently assume that protection. Conversely, a threat model that excludes original base weights must state the exclusion and explain its relevance. Moving a restriction from the attacker definition into an unstated implementation assumption overstates the conclusion.

Beyond wrappers. The criterion also applies to full-parameter alignment, unlearning, circuit breakers, and pretraining filtering, using method-specific intervention budgets and behavioral outcomes. Our results do not evaluate those methods. Reports should identify checkpoints, intervention manifests, matched evaluation protocols, paired labels, and total search budgets. The supplement supports aggregate reanalysis, not full training or generation reproduction.

9 Conclusion

Removal-cost grounding asks whether an actor can weaken safety while preserving useful behavior under an explicit intervention budget. Its value lies in the conjunction of those requirements and in the discipline of testing the cheapest admissible route. Mycelium v0 supplies a limited but instructive case: destructive removal incurs utility loss, while a reconnectable base path offers a less destructive route with suggestive safety erosion. Corrected provenance, separate generation panels, and checkpoint-aware analysis narrow what can be concluded. Durable open-weight safety remains an empirical target; a costly ablation is insufficient evidence that it has been achieved.

Limitations

The safety panels are small, use local proxies, and lack a native generation baseline on identical prompts. We cannot reconstruct an observed paired test, annotator agreement, or cross-panel intervention equivalence. The study includes no larger established safety benchmark, cross-family bypass replication, or matched random-path control. The criterion is newly formalized and was not preregistered for the archived experiments.

The utility reanalysis removes identified sample-limit and snapshot conflicts, but archival metadata is not a complete execution trace. Some metric tokens were altered by the original anonymization process; the supplement preserves and documents this rather than inventing missing names. Task averages, relative scores, and task-bootstrap intervals

have different interpretations and do not establish deployment utility. The GPT-OSS static CE audit additionally differs from the broad removal matrix: several family-zero conditions improve utility CE. This discrepancy reinforces the need to keep intervention and metric identities separate.

The available package lacks complete final training sources, exact objective schedules, model-weight files, and raw generation annotations. Equation (5) is an audit abstraction, not a verified reimplementation of every checkpoint. The experiment family labels do not prove semantic localization. No finite unsuccessful attack search establishes a general lower bound on removal cost.

Ethics and Reproducibility Statement

This work concerns defensive evaluation of post-release model integrity. We report aggregate measurements without raw harmful requests, harmful completions, or runnable attack checkpoints. The supplementary package contains sanitized score records, source hashes, analysis code, and a claim-to-evidence map. It reproduces the statistical reanalysis from supplied records; it does not regenerate private model outputs. An absence of unsafe outputs after destructive removal must not be interpreted as deployment safety. Language-model assistance was used in revising prose and preparing analysis code; the empirical observations remain the archived experimental records.

References

- Andy Ardit, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. 2024. [Refusal in Language Models Is Mediated by a Single Direction](#). *Preprint*, arXiv:2406.11717.
- Jacob Austin, Augustus Odena, Maxwell Nye, Maarten Bosma, Henryk Michalewski, David Dohan, Ellen Jiang, Carrie Cai, Michael Terry, Quoc Le, and Charles Sutton. 2021. [Program Synthesis with Large Language Models](#). *Preprint*, arXiv:2108.07732.
- Stella Biderman, Hailey Schoelkopf, Lintang Sutawika, Leo Gao, Jonathan Tow, Baber Abbasi, Alham Fikri Aji, Pawan Sasanka Ammanamanchi, Sidney Black, Jordan Clive, Anthony DiPofi, Julien Etzaniz, Benjamin Fattori, Jessica Zosa Forde, Charles Foster, Jeffrey Hsu, Mimansa Jaiswal, Wilson Y. Lee, Haonan Li, and 11 others. 2024. [Lessons from the Trenches on Reproducible Evaluation of Language Models](#). *Preprint*, arXiv:2405.14782.
- Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. 2019. [PIQA: Reasoning about Physical Commonsense in Natural Language](#). *Preprint*, arXiv:1911.11641.
- Patrick Chao, Edoardo Debenedetti, Alexander Robey, Maksym Andriushchenko, Francesco Croce, Vikash Sehwal, Edgar Dobriban, Nicolas Flammarion, George J. Pappas, Florian Tramer, Hamed Hassani, and Eric Wong. 2024. [JailbreakBench: An Open Robustness Benchmark for Jailbreaking Large Language Models](#). *Preprint*, arXiv:2404.01318.
- Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, and 39 others. 2021. [Evaluating Large Language Models Trained on Code](#). *Preprint*, arXiv:2107.03374.
- Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. 2019. [BoolQ: Exploring the Surprising Difficulty of Natural Yes/No Questions](#). *Preprint*, arXiv:1905.10044.
- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. 2018. [Think you have Solved Question Answering? Try ARC, the AI2 Reasoning Challenge](#). *Preprint*, arXiv:1803.05457.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. 2021. [Training Verifiers to Solve Math Word Problems](#). *Preprint*, arXiv:2110.14168.
- Justin Cui, Wei-Lin Chiang, Ion Stoica, and Chojui Hsieh. 2025. [OR-Bench: An Over-Refusal Benchmark for Large Language Models](#). *Preprint*, arXiv:2405.20947.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. 2020. [Measuring Massive Multitask Language Understanding](#). *Preprint*, arXiv:2009.03300.
- Saad Hossain, Tom Tseng, Punya Syon Pandey, Samanvay Vajpayee, Matthew Kowal, Nayeema Nonta, Samuel Simko, Stephen Casper, Zhijing Jin, Kellin Pelrine, and Sirisha Rambhatla. 2026. [Tamper-Bench: Systematically Stress-Testing LLM Safety Under Fine-Tuning and Tampering](#). *Preprint*, arXiv:2602.06911.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2021. [LoRA: Low-Rank Adaptation of Large Language Models](#). *Preprint*, arXiv:2106.09685.
- Nathaniel Li, Alexander Pan, Anjali Gopal, Summer Yue, Daniel Berrios, Alice Gatti, Justin D. Li, Ann-Kathrin Dombrowski, Shashwat Goel, Long Phan, Gabriel Mukobi, Nathan Helm-Burger, Rassin Lababidi, Lennart Justen, Andrew B. Liu, Michael Chen, Isabelle Barrass, Oliver Zhang, Xiaoyuan Zhu, and 38 others. 2024. [The WMDP Benchmark: Measuring and Reducing Malicious Use With Unlearning](#). *Preprint*, arXiv:2403.03218.
- Stephanie Lin, Jacob Hilton, and Owain Evans. 2021. [TruthfulQA: Measuring How Models Mimic Human Falsehoods](#). *Preprint*, arXiv:2109.07958.
- Mantas Mazeika, Long Phan, Xuwang Yin, Andy Zou, Zifan Wang, Norman Mu, Elham Sakhaee, Nathaniel Li, Steven Basart, Bo Li, David Forsyth, and Dan Hendrycks. 2024. [HarmBench: A Standardized Evaluation Framework for Automated Red Teaming and Robust Refusal](#). *Preprint*, arXiv:2402.04249.
- Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. 2018. [Can a Suit of Armor Conduct Electricity? A New Dataset for Open Book Question Answering](#). *Preprint*, arXiv:1809.02789.
- Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. 2019. [WinoGrande: An Adversarial Winograd Schema Challenge at Scale](#). *Preprint*, arXiv:1907.10641.
- Maarten Sap, Hannah Rashkin, Derek Chen, Ronan LeBras, and Yejin Choi. 2019. [SocialIQA: Commonsense Reasoning about Social Interactions](#). *Preprint*, arXiv:1904.09728.
- Alexandra Souly, Qingyuan Lu, Dillon Bowen, Tu Trinh, Elvis Hsieh, Sana Pandey, Pieter Abbeel, Justin Svegliato, Scott Emmons, Olivia Watkins, and Sam Toyer. 2024. [A StrongREJECT for Empty Jailbreaks](#). *Preprint*, arXiv:2402.10260.

Mirac Suzgun, Nathan Scales, Nathanael Schärli, Sebastian Gehrmann, Yi Tay, Hyung Won Chung, Aakanksha Chowdhery, Quoc V. Le, Ed H. Chi, Denny Zhou, and Jason Wei. 2022. [Challenging BIG-Bench Tasks and Whether Chain-of-Thought Can Solve Them](#). *Preprint*, arXiv:2210.09261.

Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. 2024. [MMLU-Pro: A More Robust and Challenging Multi-Task Language Understanding Benchmark](#). *Preprint*, arXiv:2406.01574.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. 2019. [HellaSwag: Can a Machine Really Finish Your Sentence?](#) *Preprint*, arXiv:1905.07830.

Andy Zou, Long Phan, Justin Wang, Derek Duenas, Maxwell Lin, Maksym Andriushchenko, Rowan Wang, Zico Kolter, Matt Fredrikson, and Dan Hendrycks. 2024. [Improving Alignment and Robustness with Circuit Breakers](#). *Preprint*, arXiv:2406.04313.

A Task Coverage and Snapshot Corrections

The original 16-task ledger includes one GPQA entry per model and condition. Removing the 12 GPQA error-only cells leaves 180 intended cells over the 15 paper tasks: 170 score-present cells, four additional error-only cells, and six missing cells. Score presence yields 55 complete triplets. The three provenance exclusions reduce the primary analysis to 52 triplets. All records are retained in the supplementary data, including excluded scores, with their status documented.

The common ten-task set is MMLU, MBPP, Hel-laSwag, ARC Challenge, ARC Easy, BoolQ, PIQA, Social IQA, TruthfulQA MC2, and WinoGrande. We preserve the paper-table task alias selected by the archived builder. Social IQA has multiple historical loader aliases; they are not counted as separate tasks. BBH and MMLU task aggregates are not supplemented with their subtasks in the same macro-average, avoiding double counting.

A.1 Benchmark sources and metric identity

Table 10 links the 15-task comparison universe to its benchmark sources. Metric and filter strings are those retained in the archived records. The sources identify the tasks, not a guarantee that local execution reproduced every recommended benchmark protocol. In particular, a redacted HumanEval metric cannot be reconstructed from its numerical values alone.

B Implementation Information and Its Scope

Family	Archived configuration
Mistral-7B	Last 10 layers; 30 projections; rank 64; $\rho = .22$. Attention output; MLP down/up.
Qwen-35B	Last 12 layers; 12 projections; rank 32; $\rho = .18$. Linear-attention output; final attention output.
Gemma-31B	Last 12 layers; 48 projections; rank 32; $\rho = .18$. MLP gate/down/up; attention output.
GPT-OSS-20B	Layers 12–19; 32 projections; rank 32; $\rho = .18$. Attention output, key, query, value.

Table 11: Archived implementation card. Counts denote patched modules, not scalar parameters. The source-level definition of ρ is incomplete; it is not equated with α_a in Eq. (5).

The original provenance manifest includes aligned and modified variants of several model fam-

ilies. A file’s presence in that manifest does not establish its use in any particular table. Per-task metadata is the more specific evidence for benchmark execution. The revised package preserves the corresponding sanitized paths and configuration fields without distributing model weights or download instructions.

Historical training reports describe different intermediate models. For example, one Mistral training report records six patched modules and 1,966,080 trainable anchor parameters, whereas the broad benchmark wrapper metadata records 30 patched projections. Treating the former report as the exact training configuration of the latter would be unsupported. We retain this distinction instead of filling the final checkpoint’s missing loss schedule from an earlier run.

C Additional Formal Details

Gradient accounting. For one patched projection with fixed input and gate values, a loss L back-propagates through the active low-rank paths. Writing $v = \partial L / \partial y$ and $q_f = m_{a,f} g_f(x)$ gives the local derivatives

$$\begin{aligned} \frac{\partial L}{\partial B_f} &= q_f v (A_f x)^\top, \\ \frac{\partial L}{\partial A_f} &= q_f B_f^\top v x^\top. \end{aligned} \quad (6)$$

Freezing W_0 sets its optimizer update $\Delta W_0 = 0$; it does not generally make its formal derivative zero. A trainable router adds derivatives through g_f . These equations clarify the abstraction’s gradient flow; they do not establish which historical run froze its router. Exact training reproduction requires source, optimizer configuration, data manifests, and checkpoint hashes linked to each table.

Threshold sensitivity. Thresholds depend on use, rather than on the observed best result. As an illustrative reporting grid, an auditor could vary safety erosion over $\delta \in \{0.10, 0.25, 0.50\}$ and per-task utility tolerance over $\epsilon_t \in \{0.01, 0.03, 0.05\}$, while declaring coherence, helpfulness, and false-refusal requirements. This grid is a proposal, not a preregistered criterion applied to our archived experiments. We do not choose a permissive threshold after inspecting the data and label the prototype safe or unsafe on that basis.

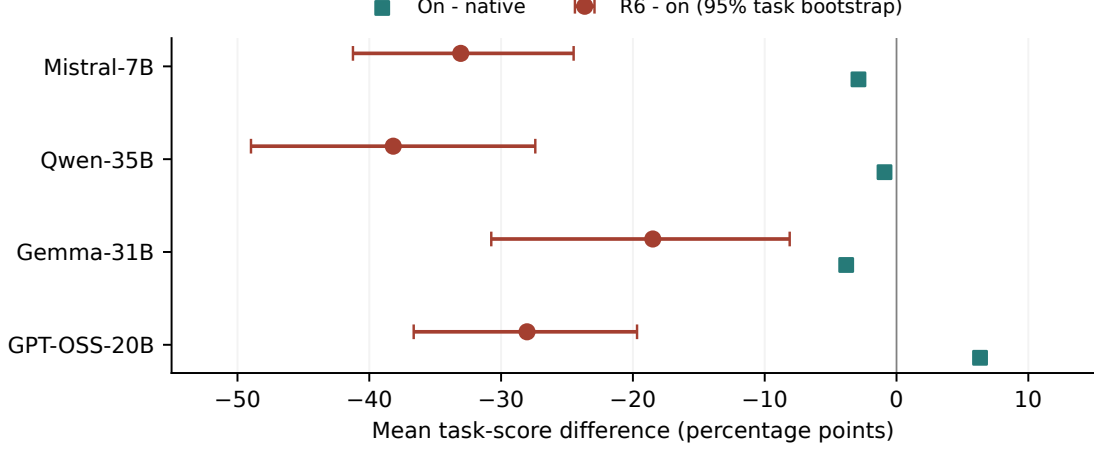


Figure 2: Corrected utility reanalysis. The removal series is the archived Phase-6 wrapper removal R_6 , not the Mistral adapter-only control. Intervals describe variation across retained task scores; they do not measure variation across training seeds or certify safety.

Family	Original incomplete triplets	Additional primary exclusions
Mistral	None among the 15 tasks	GSM8K: native/on limit 8 versus removed 1,319. OpenBookQA: removed JSON limit 4 and conflict with archived aggregate.
Qwen	BBH removed missing; MMLU-Pro missing	GSM8K: native limit 4 versus full other conditions.
Gemma	Native HumanEval runtime error; MMLU-Pro configuration errors	None identified beyond original incomplete triplets.
GPT-OSS	MMLU-Pro native and removed missing	None identified beyond original incomplete triplet.

Table 7: Coverage ledger. No unavailable score is interpreted as zero. A recorded zero in an otherwise successful task result remains a score.

D Uncertainty and Reanalysis Details

For k successes in n trials, the Wilson interval used here has center $(\hat{p} + z^2/(2n))/(1 + z^2/n)$ and half-width

$$\frac{z}{1 + z^2/n} \sqrt{\frac{\hat{p}(1 - \hat{p})}{n} + \frac{z^2}{4n^2}}, \quad (7)$$

where $\hat{p} = k/n$ and $z = 1.9599639845$. These are marginal descriptive binomial intervals. They do not account for selection of local prompts, scorer error, shared templates, or multiple comparisons.

The Panel A pairing sensitivity conditions on identical prompts and two possible joint-positive counts, zero and one. With zero joint positives, discordant counts are one and seven, giving $2(\binom{8}{0} + \binom{8}{1})/2^8 = 0.0703125$. With one joint positive, counts are zero and six, giving $2/2^6 = 0.03125$. Neither contingency table is known to be the observed one.

For the missing-cell sensitivity, let n be the retained count and $D = \sum_t (u_t(R_6) - u_t(M_0))$. Over all 15 intended tasks, arbitrary bounded ex-

cluded differences imply

$$\frac{D - (15 - n)}{15} \leq \bar{\Delta}_{15} \leq \frac{D + (15 - n)}{15}. \quad (8)$$

This calculation deliberately ignores partially observed excluded cells, yielding conservative bounds. It concerns only the bounded task-score estimand. Leave-one-task-out mean ranges are also provided in the machine-readable summary.

E Supplementary Reproduction

Run `python reproduce.py` from the supplementary directory. NumPy reproduces tables, task-bootstrap intervals, relative-loss statistics, and missing-cell bounds. Matplotlib additionally renders the two figures. No remote model service or GPU is used. Input hashes, metric fields, and retained configuration metadata link the reanalysis to the supplied benchmark records. The README identifies which historical panels can be recomputed and which remain paper-reported aggregate evidence.

Task	Mistral			Qwen		
	Native	On	R_6	Native	On	R_6
BBH	0.564	0.511	0.000	0.156	0.703	—
MMLU	0.598	0.592	0.229	0.838	0.824	0.229
MMLU-Pro	0.361	0.295	0.000	—	—	—
GSM8K	0.375 [†]	0.375 [†]	0.000 [†]	0.500 [†]	0.276 [†]	0.000 [†]
HumanEval	0.067	0.000	0.000	0.604	0.543	0.000
MBPP	0.394	0.364	0.000	0.660	0.698	0.000
HellaSwag	0.833	0.809	0.256	0.834	0.791	0.264
ARC Challenge	0.599	0.542	0.290	0.552	0.510	0.260
ARC Easy	0.817	0.779	0.263	0.720	0.744	0.248
BoolQ	0.864	0.853	0.378	0.851	0.883	0.378
OpenBookQA	0.488 [†]	0.472 [†]	0.500 [†]	0.442	0.446	0.290
PIQA	0.838	0.819	0.516	0.830	0.811	0.532
Social IQA	0.658	0.654	0.353	0.590	0.571	0.336
TruthfulQA MC2	0.593	0.593	0.480	0.547	0.551	0.480
WinoGrande	0.747	0.748	0.496	0.739	0.725	0.496

Table 8: Per-task scores from supplied JSON records. Daggered triplets are excluded from the primary summary. Mistral OpenBookQA $R_6 = 0.500$ is a four-example conflicting snapshot, not the archived aggregate score of 0.266. Missing values are shown as dashes.

Task	Gemma			GPT-OSS		
	Native	On	R_6	Native	On	R_6
BBH	0.812	0.719	0.000	0.000	0.345	0.000
MMLU	0.470	0.459	0.229	0.564	0.670	0.229
MMLU-Pro	—	—	—	—	0.376	—
GSM8K	0.702	0.438	0.000	0.251	0.247	0.000
HumanEval	—	0.000	0.000	0.268	0.579	0.000
MBPP	0.460	0.378	0.000	0.000	0.014	0.000
HellaSwag	0.396	0.372	0.264	0.577	0.632	0.263
ARC Challenge	0.243	0.247	0.252	0.484	0.477	0.256
ARC Easy	0.324	0.319	0.258	0.757	0.724	0.251
BoolQ	0.707	0.671	0.378	0.755	0.816	0.378
OpenBookQA	0.288	0.286	0.308	0.400	0.424	0.298
PIQA	0.566	0.569	0.519	0.770	0.751	0.529
Social IQA	0.406	0.402	0.335	0.529	0.595	0.334
TruthfulQA MC2	0.514	0.537	0.475	0.532	0.526	0.484
WinoGrande	0.526	0.520	0.496	0.669	0.642	0.496

Table 9: Per-task scores for Gemma and GPT-OSS. Missing or error-only cells are shown as dashes and excluded from complete-triplet summaries. Recorded zeros remain observed scores.

Task	Source	Archived metric and filter
BBH	Suzgun et al. (2022)	<code>exact_match, get-answer</code>
MMLU	Hendrycks et al. (2020)	<code>acc, none</code>
MMLU-Pro	Wang et al. (2024)	<code>exact_match, custom-extract</code>
GSM8K	Cobbe et al. (2021)	<code>exact_match, strict-match</code>
HumanEval	Chen et al. (2021)	<code>Redacted token, create_test filter</code>
MBPP	Austin et al. (2021)	<code>pass_at_1, none</code>
HellaSwag	Zellers et al. (2019)	<code>acc_norm, none</code>
ARC Challenge / Easy	Clark et al. (2018)	<code>acc_norm, none</code>
BoolQ	Clark et al. (2019)	<code>acc, none</code>
OpenBookQA	Mihaylov et al. (2018)	<code>acc_norm, none</code>
PIQA	Bisk et al. (2019)	<code>acc_norm, none</code>
Social IQA	Sap et al. (2019)	<code>acc, none</code>
TruthfulQA MC2	Lin et al. (2021)	<code>acc, none</code>
WinoGrande	Sakaguchi et al. (2019)	<code>acc, none</code>

Table 10: Benchmark sources and preserved metric identifiers. HumanEval’s metric token was replaced upstream by an anonymization placeholder; its exact original spelling is unavailable. Metric strings are retained without silently renaming them.